CareProof Agent: Verified-State Control for Reliable Healthcare Workflows

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Abstract

χ-Bench evaluates long-horizon agents on simulated provider prior authorization, payer utilization management, and care management. Its strict binary reward makes near-complete workflows equivalent to failures. CareProof is a reliability-focused control protocol integrated with the official OpenAI Agents harness. It constrains scope, separates evidence from unknowns, orders dependencies, verifies mutations, and requires proof of terminal state. The method contains no task-specific answers or protected handbook content and is intended to generalize to private tasks.

1. Problem

Healthcare-administration workflows combine clinical evidence, policy, operational state machines, structured artifacts, and conversations. A capable language model can still fail by skipping one prerequisite, acting outside its role, duplicating a record, inferring consent, or stopping after an intermediate action. Under χ-Bench's all-checks-must-pass reward, these control failures dominate plausible but incomplete answers.

2. Design objectives

CareProof targets four objectives: (1) procedural completeness, (2) evidence fidelity, (3) mutation safety, and (4) reproducible final-state verification. Efficiency remains secondary to correctness, although the protocol explicitly discourages unnecessary calls and duplicate creation.

3. Reliability protocol

The controller first extracts permitted namespaces, target identity, terminal objective, and prohibitions. It then reads the authorized handbook index and task tool reference. A private evidence ledger records sourced facts, missing facts, and conflicts. Before mutation, the agent inspects current state and exact tool schemas. Actions proceed in dependency order. Structured errors are interpreted before an idempotent retry. Every consequential mutation is followed by state verification. The agent stops only after auditing required artifacts, fields, attachments, routing, status, and follow-up ownership.

4. Conversation controls

Care-management and peer-to-peer tasks require more than database updates. CareProof treats identity checks, consent, disclosures, teach-back, refusal, escalation, and closing as performed transcript events. Consent is never inferred from cooperation. Unsupported clinical or policy facts are prohibited, and refusals follow the authorized workflow rather than being optimized away.

5. Implementation

The implementation appends a dedicated reliability protocol to the official OpenAI Agents system prompt. The submission uses the official MCP connection, local tool set, retry pipeline, logging, trajectory export, and submission lifecycle. The frozen configuration identifies openai/gpt-5.6-sol, one attempt per public task, the chi-bench-v1.0.0 dataset revision, and all three required domains.

6. Threat model

The protocol addresses premature completion, unsupported evidence, wrong-role tools, duplicate mutations, silent partial failures, conversational shortcutting, and final state drift. It does not attempt to access hidden expectations, simulator source, solutions, private tasks, or verifier-only artifacts. Protected handbook material is loaded only at authorized runtime and is never embedded in source or reports.

7. Verification

Two targeted tests assert that evidence, namespace, consent, simulator-boundary, and post-mutation verification instructions remain present. The full unit suite completed with 264 passing tests and 6 environment-dependent skips. The official submission schema validates. These results demonstrate integration correctness only; they are not substituted for official benchmark evaluation.

8. Evaluation policy

No public pass@1 or pass^3 value is claimed without an authorized, complete run and unaltered verifier evidence. The official run requires approved access to the gated Managed-Care Operations Handbook, the pinned public dataset, an agent-model credential, and the official judge/simulator credential. Each CSV reward must be copied from the corresponding official verifier result: 1 only for a verified pass, otherwise 0.

9. Reproducibility

The package includes source, prompt, tests, the official dependency lock, a frozen submission YAML, exact model routing, setup commands, and an evidence policy. Secrets are supplied only through private environment variables. The same frozen system must be used for all private tasks.

10. Limitations

A prompt-level controller cannot guarantee compliance, and LLM behavior remains nondeterministic. Extra verification can increase latency and tool use. Correctness depends on model capability, tool contracts, simulator availability, and appropriate handbook retrieval. The current submission has not claimed official task results before gated access and evaluation are complete.

11. Ethics and intended use

CareProof is a research artifact for synthetic χ-Bench cases. It is not a medical device, clinical decision system, or production payer/provider automation product. No real patient data are used. Consequential real-world healthcare decisions require qualified human oversight and appropriate legal, clinical, privacy, and security review.

12. Conclusion

CareProof reframes healthcare-agent performance as verified state transition rather than answer generation. By encoding scope, provenance, dependency, mutation, conversation, and terminal-state controls in one reusable protocol, the submission targets recurring failure classes while preserving compatibility with the official χ-Bench runtime.

References

[1] H. Chen et al., “CHI-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?” arXiv:2605.16679, 2026.
[2] actAVA AI, “χ-Bench,” official repository and competition documentation, 2026.
[3] OpenAI, “OpenAI Agents SDK,” software documentation, 2026.